



Cohort canonical evaluation — 29 agents

idea-01-lateral-attribution · 130 held-out segments · V0
yaw 0.01456 rad/s, CTE 147.44 m · generated
2026-05-31 00:31

- Executive summary

AGENTS (OK / TOTAL)

28 / 29

EVAL POOL

130 segs

V0 YAW RMSE

0.01456

V0 CTE RMSE

147.44 m

WALL TIME

4.19s

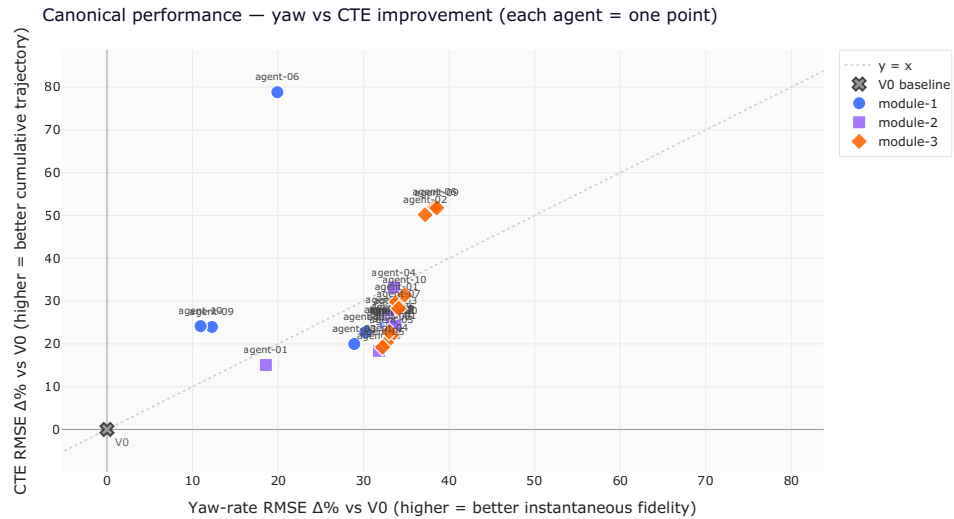
Best yaw: m3-agent-09 **+38.6%** · Best CTE: m1-agent-06

+78.8%

Winning both KPIs $\geq +30\%$ (5 agents): m3-agent-09, m3-agent-06, m3-agent-02, m3-agent-10, m2-agent-04

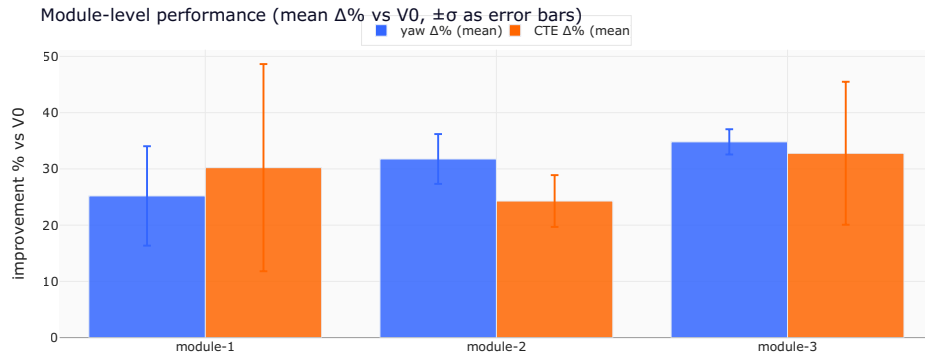
1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is $y = x$: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



2. Module-level aggregation

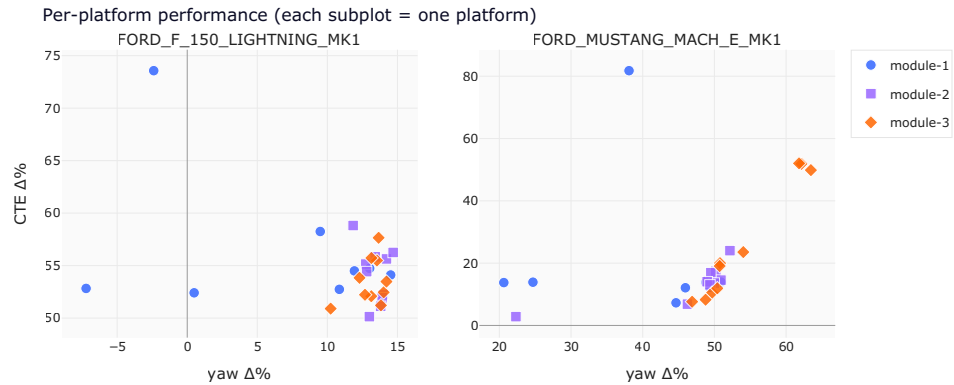
Performance grouped by which module shipped the agent.
Mean $\Delta\%$ over only the agents that reconstructed successfully;
failures shown separately.



family	n ok / total	yaw $\Delta\%$ (mean $\pm \sigma$)	CTE $\Delta\%$ (mean $\pm \sigma$)
module-1	8/9	+25.2% $\pm$ 8.8%	+30.2% $\pm$ 9.4%
module-2	10/10	+31.8% $\pm$ 4.4%	+24.3% $\pm$ 4.5%
module-3	10/10	+34.8% $\pm$ 2.2%	+32.8% $\pm$ 12.8%

• 3. Per-platform breakdown

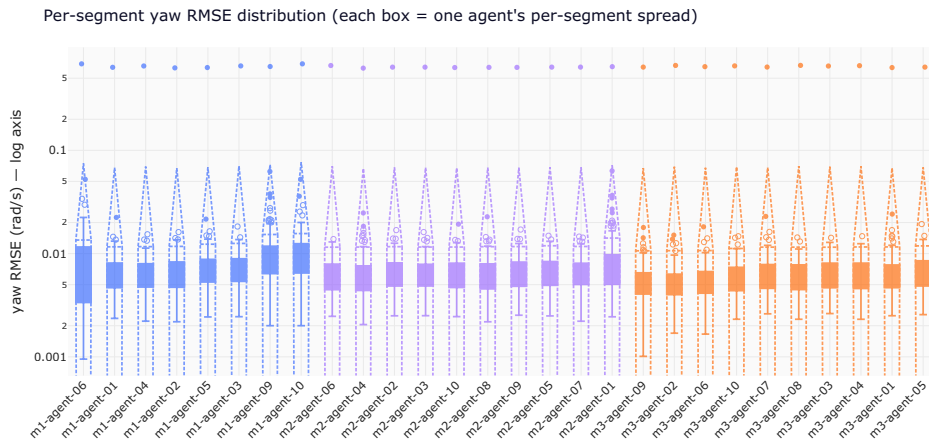
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.



platform	agents	yaw Δ% (mean $\pm$ σ)
FORD_F_150_LIGHTNING_MK1	28	+11.3% $\pm$ 5.2%
FORD_MUSTANG_MACH_E_MK1	28	+47.5% $\pm$ 10.0%

- 4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

agent	family	status	yaw V0	yaw
m1-agent-01	module-1	ok	0.014563	0.00
m1-agent-02	module-1	ok	0.014563	0.00
m1-agent-03	module-1	ok	0.014563	0.01
	module-1	ok	0.014563	0.00

agent	family	status	yaw V0	yaw
m1-agent-04				
m1-agent-05	module-1	ok	0.014563	0.01
m1-agent-06	module-1	ok	0.014563	0.01
m1-agent-07	module-1	import_failed	—	
m1-agent-09	module-1	ok	0.014563	0.01
m1-agent-10	module-1	ok	0.014563	0.01
m2-agent-01	module-2	ok	0.014563	0.01
m2-agent-02	module-2	ok	0.014563	0.00
m2-agent-03	module-2	ok	0.014563	0.00
m2-agent-04	module-2	ok	0.014563	0.00
m2-agent-05	module-2	ok	0.014563	0.00
m2-agent-06	module-2	ok	0.014563	0.00
	module-2	ok	0.014563	0.00

agent	family	status	yaw V0	yaw
m2-agent-07				
m2-agent-08	module-2	ok	0.014563	0.00
m2-agent-09	module-2	ok	0.014563	0.00
m2-agent-10	module-2	ok	0.014563	0.00
m3-agent-01	module-3	ok	0.014563	0.00
m3-agent-02	module-3	ok	0.014563	0.00
m3-agent-03	module-3	ok	0.014563	0.00
m3-agent-04	module-3	ok	0.014563	0.00
m3-agent-05	module-3	ok	0.014563	0.00
m3-agent-06	module-3	ok	0.014563	0.00
m3-agent-07	module-3	ok	0.014563	0.00
m3-agent-08	module-3	ok	0.014563	0.00
	module-3	ok	0.014563	0.00

agent	family	status	yaw V0	yaw
m3-agent-09				
m3-agent-10	module-3	ok	0.014563	0.00

- **6. Calibration cards — fitted coefficients**

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "s_scale": 0.94203,
    "delta_offset_rad": 0.00102,
    "K_us": 0.00304,
    "tau_yaw_s": 0.04592
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "s_scale": 1.21716,
    "delta_offset_rad": -5e-05,
    "K_us": 0.00304,
    "tau_yaw_s": 0.05262
  },
}
```

```
"TESLA_MODEL_3": {
  "L": 2.875,
  "s_scale": 1.0,
  "delta_offset_rad": 0.0,
  "K_us": 0.0025,
  "tau_yaw_s": 0.05
},
"_method": "Fit on 60% of segments per platform; model: yr_ss =
"_tesla_note": "TESLA_MODEL_3 has no measured yaw-rate truth; co
}
```

m1-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "a": 0.9127536047016024,
    "b": 0.0007692876565060736,
    "delta_off": 0.0012208376981866274,
    "tau": 0.06378078127896805
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "a": 1.1599450659882828,
    "b": 0.0008023134078587542,
    "delta_off": 1.3682104928342577e-05,
    "tau": 0.06877523527457137
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "a": 1.0,
    "b": 0.0,
    "delta_off": 0.0,
    "tau": 0.0
  }
}
```

m2-agent-01

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K_us": 0.0045433459530302405,
    "delta0": 0.0013939755442043738,
    "tau": 0.059696812148648346
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K_us": 0.0008613349618482884,
    "delta0": -2.4463406949899732e-05,
    "tau": 0.058058773969275335
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0007,
    "delta0": 0.0,
    "tau": 0.08
  }
}
```

m2-agent-03

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.7874160302267805,
    "K": 0.0010601777605335054,
    "d0": 0.0012897224766212265,
    "tau": 0.05
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.4765476192620257,
    "K": 0.0009714616153136761,
    "d0": 4.480994570776638e-05,
    "tau": 0.05
  }
}
```

m2-agent-05

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "g": 0.978386024867586,
    "K_us": 0.003952369466136279,
    "delta0": 0.0013386063106657791,
    "tau": 0.05245353568404573
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "g": 1.212362348609587,
    "K_us": 0.0030155666104823165,
    "delta0": -0.00017873648699720767,
    "tau": 0.05847378305110825
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "g": 1.0,
    "K_us": 0.003,
```

```
"delta0": 0.0,  
  "tau": 0.05  
}  
}
```

m2-agent-09

```
{  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "s": 1.203330959234065,  
    "K": 0.0029361363287092115,  
    "bias": 0.00021448227575312877,  
    "tau": 0.032147914442713796,  
    "delay": 1  
  },  
  "FORD_F_150_LIGHTNING_MK1": {  
    "s": 0.9772602779030508,  
    "K": 0.003915796445879638,  
    "bias": -0.004423910419199254,  
    "tau": 0.022540457646443024,  
    "delay": 1  
  }  
}
```

m3-agent-01

```
{  
  "_doc": "Per-platform fitted coefficients for the kinematic-bicycle model",  
  "tau_s": 0.08,  
  "FORD_F_150_LIGHTNING_MK1": {  
    "g": 0.9567028992805204,  
    "K_us": 0.0033801343956829,  
    "delta_0": 0.0008461582112823263,  
    "tau": 0.032147914442713796,  
    "bias": 0.00021448227575312877,  
    "K": 0.0029361363287092115,  
    "s": 1.203330959234065,  
    "delay": 1  
  }  
}
```

```
    "L": 3.7
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.1725815243702862,
    "K_us": 0.0025152837759692456,
    "delta_0": 0.00013949413303459602,
    "L": 2.984
  }
}
```

m3-agent-03

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g0": 0.9224096029662896,
    "g2": 0.15876361430939134,
    "delta0": -0.0010913875270846236,
    "L_eff": 3.6413019931403277,
    "K0": 0.004767766094094899,
    "K1": -5.858241730532427e-05,
    "tau": 0.0740452029657968
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g0": 1.0219456519707983,
    "g2": 0.40092616819123517,
    "delta0": 5.9966690969289335e-05,
    "L_eff": 2.771534314392467,
    "K0": 0.0015320260055171069,
    "K1": 1.5004138297234027e-05,
    "tau": 0.08533887973933936
  }
}
```

m3-agent-05

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.1935822683980741,
    "delta_offset": 0.00016378819390629683,
    "K_us": 0.0027507391842350546,
    "tau": 0.07010533877461887,
    "L": 2.984
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.9786307745986058,
    "delta_offset": -0.0012312582055424145,
    "K_us": 0.003976165863986497,
    "tau": 0.06037312785047078,
    "L": 3.7
  }
}
```

m3-agent-07

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.1762462360941122,
    "d0": 0.00010063505427400637,
    "K_us": 0.0026382633004772204,
```

```
"tau": 0.06840716946070041,
"g_v3": 1.1770261504574182,
"d0_v3": 0.00014959445374349638,
"K_us_v3": 0.0026591289393425076,
"tau_v3": 0.06881984429222407,
"alpha_v3": 3.414239845943319e-09,
"g_v1": 1.173581598798327,
"d0_v1": 9.779075692684366e-05,
"K_us_v1": 0.0026023048615728033,
"L": 2.984
},
"FORD_F_150_LIGHTNING_MK1": {
  "g": 0.9812529513258541,
  "d0": 0.0011869609888145339,
  "K_us": 0.0041443305277957445,
  "tau": 0.06130822211279313,
  "g_v3": 0.9812464336315232,
  "d0_v3": 0.0011870345278419555,
  "K_us_v3": 0.004144503544848034,
  "tau_v3": 0.06128822752272341,
  "alpha_v3": 3.986979263377587e-09,
  "g_v1": 0.9795065634523334,
  "d0_v1": 0.0011752337517999898,
  "K_us_v1": 0.004081817473342322,
  "L": 3.7
}
}
```

m3-agent-09

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "model": "v1_global_delta0",
    "g": 0.8633847850655261,
    "delta0": 0.001333845368641501,
    "L_eff": 3.262313536845297,
    "K_us": 0.00349772209138941,
    "tau": 0.05952805407230728
  }
}
```



```
},  
"FORD_MUSTANG_MACH_E_MK1": {  
  "model": "v4_per_segment_delta0",  
  "g": 0.8907859709885533,  
  "L_eff": 2.2160488557450244,  
  "K_us": 0.002016048675027096,  
  "tau": 0.0690564359086768,  
  "delta0_fallback": -0.00010084780741412298  
}  
}
```

7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

format check	pass	fail
agent_folder_exists	29	0
has_manifest_json	29	0
manifest_parsable	29	0
manifest_declares_predict_callable	29	0

format check	pass	fail
manifest_declares_platform_support	29	0
has_predict_py	29	0
has_coeffs_json	22	7
has_report	25	4

Failure reasons across the cohort:

- import_failed — 1 agent(s)

8. Worst-of-cohort

Lowest yaw $\Delta\%$

- m1-agent-10
(+10.9%)
- m1-agent-09
(+12.3%)
- m2-agent-01
(+18.6%)

Lowest CTE $\Delta\%$

- m2-agent-01
(+15.1%)
- m2-agent-05
(+18.4%)
- m3-agent-05
(+19.2%)



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